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Explainable Federated Analytics for Multivariate Time-Series Forecasting with Personalized Drift Adaptation

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1 Problem Definition & Originality

Problem Statement:

Collaborative time-series forecasting across distributed organizations requires model training without centralizing sensitive data. Although Federated Learning enables privacy-preserving distributed training, existing systems fail to simultaneously address four critical challenges:

1. Privacy preservation,
2. Client heterogeneity (non-IID distributions),
3. Concept drift in evolving environments,
4. Explainability of predictions.

This limitation restricts trustworthy deployment in real-world domains such as healthcare analytics, smart grids, industrial IoT systems, and smart cities.

Originality Claim (Extension-Based Research):

This research extends classical Federated Averaging and Personalized Federated Learning frameworks by introducing:

- A hybrid global–personalized architecture,
- Online concept drift detection integrated within federated rounds,
- Attention-based explainable forecasting,
- Adaptive retraining triggered by statistical thresholds.

Unlike prior works that address these challenges independently, this study proposes a unified framework integrating personalization, drift adaptation, and explainability within a single federated architecture.

2 Literature Survey & Gap Analysis

Federated Learning introduced decentralized training via weighted model aggregation. Later works introduced personalization layers to mitigate client heterogeneity. Drift detection has been studied primarily in centralized machine learning systems.

However, survey findings indicate:

- Personalization improves accuracy but lacks interpretability.

- Drift detection rarely operates within federated loops.
- Explainable AI techniques are seldom integrated into federated time-series forecasting.
- Multivariate time-series under federated settings remains under-explored.
- Most studies assume static environments.

Research Gap:
There is no comprehensive federated framework that simultaneously provides:

- Personalized learning for non-IID clients,
- Online concept drift monitoring,
- Attention-based explainable predictions,
- Robust multivariate forecasting performance.

3 Proposed Methodology (Technical Knowledge)

Proposed Model: Federated Personalized Drift-Aware Attention Framework (FPDAF)
The framework consists of three integrated modules:

1. Global Shared Backbone for capturing common temporal dynamics.
2. Client-Specific Personalization Layers for local adaptation.
3. Drift-Aware Monitoring Module using statistical change detection.

Attention Mechanism:
Temporal attention weights are computed as:

$$\alpha_t = \frac{\exp(e_t)}{\sum_k \exp(e_k)}$$

where α_t represents the importance of timestep t.

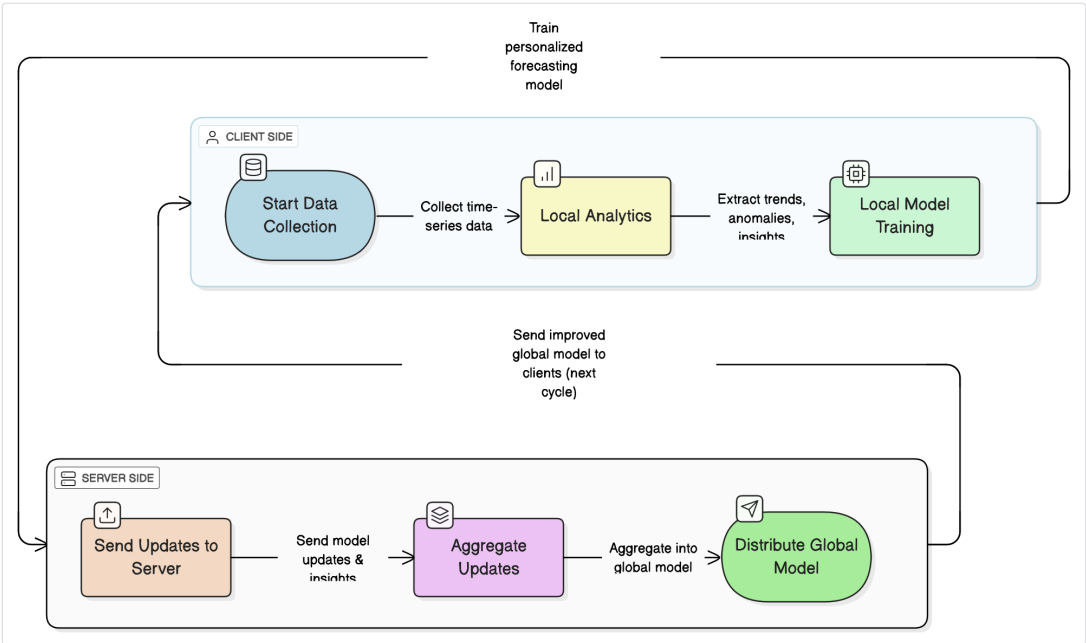

Figure 1: Adaptive Workflow Framework (AWF) illustrating federated personalization and drift adaptation pipeline

Training Flow: Local Training → Secure Upload → Global Aggregation → Model Redistribution

Federated Aggregation:

$$w^{t+1} = \sum_{i=1}^{N} \frac{n_i}{n} w_i^t$$

Concept Drift Detection (CUSUM):

$$S_t = \max(0, S_{t-1} + (x_t - \mu - k))$$

If $S_t > h$, adaptive retraining is triggered.

4 Evaluation Strategy

The framework will be evaluated using:

- Prediction metrics: RMSE, MAE, MAPE.
- Personalization gain over global FedAvg.
- Drift responsiveness after distribution shift.
- Explainability consistency via attention stability.

Baselines include:

- Centralized LSTM/Transformer.
- Standard Federated Averaging.
- Non-personalized federated models.

5 Current Status & Plan

Current Status:

- Literature survey of 25 peer-reviewed papers completed.
- Research gap formalized.
- Mathematical framework defined.
- Architecture conceptualized.

Next Steps:

- Implementation of personalized federated model.
- Integration of drift detection module.
- Multi-client simulation experiments.
- Benchmarking and ablation study.

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